

Практикум A2:archive

Twenty-seven questions, 77 marks. Every question the archive asks about geometric sequences, from November 2021 to November 2025, in the order of the seven techniques rather than the order of the papers.

О практикуме

Собран из двух тем сразу:

`number_algebra.geometric_sequences` целиком (17 блоков, 55 баллов) и остаток `number_algebra.sequences` после A1 (10 блоков, 22 балла) — те вопросы, что начинаются арифметической последовательностью и кончаются геометрической.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-a2-geometric-sequences

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

A2 archive — geometric sequences and infinite sums, all of it

Twenty-seven questions, 77 marks.

Every question the archive asks about geometric sequences, from November 2021 to November 2025, in the order of the seven techniques rather than the order of the papers.

No theory. No worked examples. Open it, answer twenty-seven questions, close it. The theory is in the practicum, `practicum-a2-geometric-sequences.ipynb`.

§	technique	questions	marks
1	The ratio and a term	5	13
2	Add the first n	3	10
3	The infinite sum	4	15
4	A series written as a Σ	3	7
5	Three quantities in a geometric sequence	6	14
6	The smallest n	3	12
7	The ratio is an expression	3	6

Answers go on one line each. The checks are the same ones the practicum uses and they store nothing: a sequence is handed over as a rule — a first term and a ratio — and the check multiplies its way along it. The infinite sum is reached by adding until the tail stops mattering, which is why a series that does not shrink cannot be summed at all.

Two questions have no cell. *State a reason why the sum does not exist and show that $k^2 - 10k - 24 = 0$ cannot be checked by anything;* what the markscheme accepts for them is written out in the solutions at the end.

Two questions start in the middle. May 2023 TZ1 Q10 and May 2025 TZ3 Q10 begin arithmetic and finish geometric. Their earlier parts are in the A1 archive, and it is said again where it happens.

Solutions are at the very bottom, deliberately far away.

```

import sys
sys.path.append('..')          # from
practicum/number_algebra to practicum/kit/
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
geometric, term, total, infinite, ...

language('en')                 # this
notebook is in English, and so are the checks

n = symbols('n')               # the index:
every one of these papers calls it n
m = symbols('m')               # a second
index, in § 5
p = symbols('p')               # the ratio
when the paper calls it p
r = symbols('r')               # the ratio
when it is the unknown
a, s, t = symbols('a s t')     # letters that
stay letters
x = symbols('x')               # the ratio is
allowed to be an expression in x
                                # k already
comes from kit

# Every sequence below may be built from an
answer you have not written
# yet. geometric() returns an empty rule in
that case, so the notebook
# runs top to bottom while it is still blank.

def ratios(rule):
    # the geometric sequence whose nth term
    is rule(n)

```

```
try:
    first, second = rule(1), rule(2)
except TypeError:
    return geometric(..., ...)
if blank(first, second) or first == 0:
    return geometric(..., ...)
return geometric(first, second / first)

print('ready; sympy', sp.__version__)
print('a geometric sequence:', geometric(3,
2), '→ term 5 is',
    term(geometric(3, 2), 5))
```

§ 1. The ratio and a term

Five questions, 13 marks. $u_n = u_1 r^{n-1}$, read forwards and backwards, with percentages counting as ratios.

1.1 — May 2025 TZ2 Paper 2 Q7(a), 3 marks

A geometric sequence has first term 80 and fourth term 0.74088. Find the second term.

```
q1_1r = ...      # the common ratio
q1_1 = ...      # the second term

mine = geometric(80, q1_1r)

verify_term('1.1 (r)', 0.74088, mine, 4)
verify_term('1.1', q1_1, mine, 2)
```

1.2 — May 2025 TZ3 Paper 1 Q10(b) (i), 3 marks

The sequence $\{u_n\}$ has first three terms $u_1 = k - 5$, $u_2 = 3 - 2k$ and $u_3 = 5k + 3$, where $k \in \mathbb{R}$. (Part (a), where the sequence is arithmetic, is in the A1 archive.)

Consider the case where $k = 12$. Show that the first three terms of $\{u_n\}$ form a geometric sequence.

```
q1_2t = [...]      # the first three terms when
k = 12
q1_2r = ...        # the common ratio

printed = [k - 5, 3 - 2 * k, 5 * k + 3]

verify_start('1.2', q1_2t, ratios(lambda i:
printed[i - 1].subs(k, 12)))
verify_geometric('1.2 geometric', q1_2t)
verify_ratio('1.2 (r)', q1_2r, q1_2t)
```

1.3 — May 2025 TZ1 Paper 1 Q5(b), 3 marks

Ten rectangular picture frames have areas $20 \left(\frac{9}{4}\right)^{n-1} \text{ cm}^2$ for $n = 1, \dots, 10$. Find the median area of the ten frames, giving your answer in the form $q \left(\frac{9}{4}\right)^4 \text{ cm}^2$, where $q \in \mathbb{Q}^+$.

```
q1_3 = ...          # q

areas = geometric(20, Rational(9, 4))
from_median = (... if blank(q1_3)
                  else total(areas, 4) + 2 *
q1_3 * Rational(9, 4) ** 4)

verify_total('1.3', from_median, areas, 6)
```


1.4 — May 2024 TZ1 Paper 2 Q1(a), 2 marks

Darren buys a car for \$35 000. The value of the car decreases by 15% in the first year. Find the value of the car at the end of the first year.

1.5 — May 2024 TZ1 Paper 2 Q1(b), 2 marks

After the first year, the value of the car decreases by 11% in each subsequent year. Find the value of Darren's car 10 years after he buys it, giving your answer to the nearest dollar.

```
q1_4 = ...          # the value after one year
q1_5 = ...          # the value after ten years,
to the nearest dollar

car = geometric(q1_4, 0.89)

verify_term('1.4', 35000 * 0.85, car, 1)
verify_term('1.5', q1_5, car, 10)
```

§ 2. Add the first n

Three questions, 10 marks. $S_n = \frac{u_1(r^n - 1)}{r - 1}$, and counting the terms before using it.

2.1 — May 2024 TZ1 Paper 1 Q5(a), 1 mark

Consider a geometric sequence with first term 1 and common ratio 10. S_n is the sum of the first n terms. Find an expression for S_n in the form $\frac{a^n - 1}{b}$, where $a, b \in \mathbb{Z}^+$.

2.2 — May 2024 TZ1 Paper 1 Q5(b), 4 marks

Hence show that $S_1 + S_2 + \cdots + S_n = \frac{10(10^n - 1) - 9n}{81}$.

Nothing to hand over at the end of a "show that" — so write down the geometric half of the working instead: the value of $10 + 100 + \cdots + 10^n$, in terms of n .

```

q2_1 = ...          # S_n, in terms of n
q2_2 = ...          # 10 + 100 + ... + 10**n, in
terms of n

verify_total('2.1', q2_1, geometric(1, 10),
n)
verify_total('2.2', q2_2, geometric(10, 10),
n)

```

2.3 — May 2025 TZ1 Paper 1 Q5(a), 5 marks

Frame F_1 has width 4 cm and height 5 cm; the width and height of F_n are each increased by 50% to give those of F_{n+1} , for $1 \leq n \leq 9$.

(i) Show that the area of frame F_n is $20 \left(\frac{9}{4}\right)^{n-1} \text{ cm}^2$. **(ii)** Hence find the mean area of the ten frames, in the form $p \left(\left(\frac{9}{4}\right)^a - 1\right) \text{ cm}^2$, where $p \in \mathbb{Q}^+$, $a \in \mathbb{Z}^+$.

```

q2_3p = ...          # p
q2_3a = ...          # a

areas = geometric(20, Rational(9, 4))
from_mean = (... if blank(q2_3p, q2_3a)
               else 10 * q2_3p * (Rational(9,
4) ** q2_3a - 1))

verify_total('2.3', from_mean, areas, 10)

```

§ 3. The infinite sum

Four questions, 15 marks. $S_\infty = \frac{u_1}{1-r}$, and the condition $|r| < 1$ that makes it exist.

3.1 — November 2021 Paper 2 Q5(b), 3 marks

The sum of the first n terms of a geometric sequence is $S_n = \sum_{i=1}^n \frac{2}{3} \left(\frac{7}{8}\right)^i$.

Find S_∞ .

```
q3_1 = ...          # the sum to infinity

verify_infinite('3.1', q3_1,
geometric(Rational(2, 3) * Rational(7, 8),
Rational(7, 8)), exact=True)
```

3.2 — May 2025 TZ3 Paper 1 Q10(b) (ii), 1 mark — no cell

The first three terms of $\{u_n\}$ with $k = 12$ are 7, -21 , 63 and the sequence is geometric. State a reason why the sum of an infinite number of terms of this sequence does not exist.

One sentence, and no cell can check a sentence. The markscheme's wording is in the solutions.

3.3 — November 2025 TZ3 Paper 2 Q10(e), 5 marks

A particle P moves so that its displacement from O at time t is $s(t) = 2^{(1-\frac{t}{5})} \sin(\frac{2\pi t}{3})$, $t \geq 0$; it passes through O every T seconds. The sequence $u_1, u_2, u_3 \dots$ of largest **distances** from O in the successive intervals $0 < t < T$, $T < t < 2T$, $2T < t < 3T \dots$ is geometric, and earlier parts give $u_1 = 1.80645$, $u_2 = 1.46729$, $u_3 = 1.19181$.

(i) Determine the value of the common ratio r . **(ii)** Calculate the total distance travelled by P if it were to continue to move in this way indefinitely.

```

q3_3r = ...      # the common ratio, to three
significant figures
q3_3d = ...      # the total distance

out_and_back = geometric(2 * 1.80645, 1.46729
/ 1.80645)      # out and back each time

verify_ratio('3.3(i)', q3_3r, [1.80645,
1.46729, 1.19181])
verify_infinite('3.3(ii)', q3_3d,
out_and_back)

```

3.4 — May 2022 TZ1 Paper 1 Q10(a), 6 marks

Consider the series $\ln x + p \ln x + \frac{1}{3} \ln x + \dots$, where $x \in \mathbb{R}$, $x > 1$ and $p \in \mathbb{R}$, $p \neq 0$. Consider the case where the series is geometric.

(i) Show that $p = \pm \frac{1}{\sqrt{3}}$. **(ii)** Hence or otherwise, show that the series is convergent. **(iii)** Given that $p > 0$ and $S_{\infty} = 3 + \sqrt{3}$, find the value of x .

Divide every term by $\ln x$ and the series becomes $1, p, \frac{1}{3}$ — which is what the check is handed. For (iii) the answer goes inside the sequence, and the $3 + \sqrt{3}$ comes from the question.

```

q3_4p = [...]      # both values of p
q3_4x = ...         # the value of x

# 1, p, 1/3 is the series with ln x divided
out
verify_root_set('3.4(i)', q3_4p,
Eq(term(geometric(1, p), 3), Rational(1, 3)),
    var=p)
verify_infinite('3.4(iii)', 3 + sqrt(3),
    geometric(... if blank(q3_4x)
else ln(q3_4x), 1 / sqrt(3)))

```

§ 4. A series written as a Σ

Three questions, 7 marks. The first term is the summand at the lower limit, and the constant in front of the bracket is not it.

4.1 — November 2021 Paper 2 Q5(a), 2 marks

The sum of the first n terms of a geometric sequence is $S_n = \sum_{i=1}^n \frac{2}{3} \left(\frac{7}{8}\right)^i$.

Find the first term of the sequence, u_1 .

```
q4_1 = ...          # the first term

mine = geometric(q4_1, Rational(7, 8))

verify_term('4.1', Rational(2, 3) *
Rational(7, 8) ** 2, mine, 2)
```

4.2 — November 2025 TZ1 Paper 2 Q6(a), 3 marks

Consider the infinite series $\sum_{k=0}^{\infty} \left(\frac{53}{1000}\right) \left(\frac{1}{100}\right)^k$. Find the exact value of S_{∞} .

4.3 — November 2025 TZ1 Paper 2

Q6(b), 2 marks

Let $0.2\overline{53}$ represent the repeating decimal $0.2535353\dots$. Use your answer from part (a) to express $0.2\overline{53}$ as a fraction whose numerator and denominator have no common factors.

```
q4_2 = ...          # the exact sum to infinity
q4_3 = ...          # the repeating decimal as a
fraction

series = geometric(Rational(53, 1000),
Rational(1, 100))

verify_infinite('4.2', q4_2, series,
exact=True)
verify_infinite('4.3', ... if blank(q4_3)
else q4_3 - Rational(1, 5),
                series, exact=True)
```

§ 5. Three quantities in a geometric sequence

Six questions, 14 marks. $s^2 = at$, and the quadratic it turns into when a letter sits inside the terms.

5.1 — May 2024 TZ2 Paper 1 Q10(b), 2 marks

Consider the geometric sequence $a, s, t \dots$, where $a, s, t \neq 0$. Show that $s^2 = at$.

Nothing to hand over — so write the third term in terms of the first two.

```
q5_1 = ...      # t, in terms of a and s  
  
verify_geometric('5.1', [a, s, q5_1])
```

5.2 — May 2023 TZ1 Paper 1 Q10(d), 3 marks

An arithmetic sequence u_n has $u_1 = 5$ and $u_6 = 15$. (*Parts (a)–(c), which establish that, are in the A1 archive.*) Consider a geometric sequence v_n , where $v_2 = u_1$ and

$v_4 = u_6$. Find the possible values of the common ratio, r .

5.3 — May 2023 TZ1 Paper 1 Q10(e), 2 marks

Given that $v_{99} < 0$, find v_5 .

```
q5_2 = [...]      # both possible values of r
q5_3n = ...        # the one that makes v99
negative
q5_3 = ...         # v5

# the sequence starts at v2, so its 3rd term
is v4
verify_root_set('5.2', q5_2,
Eq(term(geometric(5, r), 3), 15), var=r)
verify_term('5.3', q5_3, geometric(5, q5_3n),
4)
```

5.4 — May 2025 TZ3 Paper 1 Q10(c) (i), 2 marks — no cell

The sequence $\{u_n\}$ has $u_1 = k - 5$, $u_2 = 3 - 2k$, $u_3 = 5k + 3$, and is geometric for a second value of k besides 12. Show that $k^2 - 10k - 24 = 0$.

A "show that" whose whole content is the algebra between the two printed lines. The expansion is in the solutions.

**5.5 — May 2025 TZ3 Paper 1 Q10(c)
(ii), 4 marks**

Find the first three terms of $\{u_n\}$ for this second value of k .

**5.6 — May 2025 TZ3 Paper 1 Q10(c)
(iii), 1 mark**

Hence write down the value of S_{2m} , the sum of the first $2m$ terms, for this second value of k .

```
q5_5k = ...      # the second value of k
q5_5t = [...]    # the first three terms
q5_5r = ...      # the common ratio they have
q5_6 = ...       # S_2m

printed = [k - 5, 3 - 2 * k, 5 * k + 3]

verify_root_set('5.5 (k)', [12, q5_5k], Eq(k
** 2 - 10 * k - 24, 0), var=k)
verify_start('5.5', q5_5t,
              ratios(lambda i: printed[i -
1].subs(k, q5_5k)
                    if not blank(q5_5k) else
...))
verify_ratio('5.5 (r)', q5_5r, q5_5t)
verify_total('5.6', q5_6, geometric(q5_5t[0],
q5_5r), 2 * m)
```

§ 6. The smallest n

Three questions, 12 marks. A table of values, a sketch or logarithms — the markscheme lists all three as equals — and then rounding the right way.

6.1 — *November 2022 Paper 2 Q3, 5 marks*

A geometric sequence has a first term of 50 and a fourth term of 86.4. The sum of the first n terms is S_n . Find the smallest value of n such that $S_n > 33\,500$.

```
q6_1r = ...      # the common ratio
q6_1 = ...      # the smallest n

mine = geometric(50, q6_1r)

verify_term('6.1 (r)', 86.4, mine, 4)
verify_least('6.1', q6_1, mine, lambda
partial: partial > 33500)
```

6.2 — *November 2021 Paper 2 Q5(c), 4 marks*

For the series $S_n = \sum_{i=1}^n \frac{2}{3} \left(\frac{7}{8}\right)^i$, find the least value of n such that $S_\infty - S_n < 0.001$.

Your own S_∞ from 3.1 is what the condition is built out of, so a wrong answer there costs you that answer only.

```
q6_2 = ...          # the least n

mine = geometric(Rational(2, 3) * Rational(7,
8), Rational(7, 8))

verify_least('6.2', q6_2, mine,
             lambda partial: q3_1 - partial <
Rational(1, 1000))
```

6.3 — May 2024 TZ1 Paper 2 Q1(c), 3 marks

Darren's car is worth \$29 750 after one year and loses 11% of its value each year after that. When he has owned it for n complete years its value is less than 10% of the original \$35 000. Find the least value of n .

```
q6_3 = ...          # the least number of
complete years

car = geometric(29750, 0.89)

verify_least('6.3', q6_3, car, lambda value:
value < 3500, what='term')
```

§ 7. The ratio is an expression

Three questions, 6 marks. The same three formulas with r that is not a number, and the convergence condition written as a set of x .

7.1 — November 2022 Paper 3 Q1(e), 2 marks

By considering $f(x) = 1 + x + x^2 + \dots + x^n$ as a geometric series, for $x \neq 1$, show that $f(x) = \frac{x^{n+1} - 1}{x - 1}$.

*One of the two marks is **R1** for a clear indication of how many terms there are, so write that down as well.*

```
q7_1n = ...      # how many terms, in terms
of n
q7_1 = ...       # the sum of that many terms

verify_total('7.1', q7_1, geometric(1, x),
q7_1n,
                var=x, values=(2, 3, Rational(1,
2)))
```

7.2 — November 2025 TZ3 Paper 3

Q2(a), 2 marks

Given $|x| < 1$, find the sum to infinity of the geometric series $1 - x^2 + x^4 - x^6 + \dots$

7.3 — May 2025 TZ1 Paper 2 Q12(b), 2 marks

Consider $f_n(x) = \sum_{r=0}^n (-2x^2)^r$. Given that $f_3(x) = 1 - 2x^2 + 4x^4 - 8x^6$, write down a similar expression for $f_4(x)$ in ascending powers of x .

```
q7_2 = ...      # the sum to infinity of 1 -
x^2 + x^4 - ...
q7_3 = ...      # f4(x)

verify_infinite('7.2', q7_2, geometric(1, -x
** 2),
                var=x, values=(Rational(1,
2), Rational(1, 3)))
verify_total('7.3', q7_3, geometric(1, -2 * x
** 2), 5,
            var=x, values=(2, Rational(1,
3)))
```



Solutions

1.1 $80r^3 = 0.74088$, so $r^3 = 0.0092610$ and $r = 0.21$ — exact, since $0.21^3 = 0.009261$. Then $u_2 = 80(0.21) = 16.8$.

1.2 With $k = 12$ the terms are 7, -21 , 63, and

$$\frac{-21}{7} = -3 = \frac{63}{-21}$$

Equal ratios, so geometric, with $r = -3$. Showing only one of the two divisions is not showing it.

1.3 The median of ten values is the mean of the 5th and 6th:

$$\frac{20 \left(\frac{9}{4}\right)^4 + 20 \left(\frac{9}{4}\right)^5}{2} = \frac{20 \left(\frac{9}{4}\right)^4 \left(1 + \frac{9}{4}\right)}{2} = \frac{65}{2} \left(\frac{9}{4}\right)^4$$

1.4 $0.85 \times 35\,000 = \$29\,750$.

1.5 29 750 is the value after **one** year, so ten years is nine more steps: $29\,750 \times$

$0.89^9 = \boxed{\$10\,423}$. Using 0.89^{10} gives \$9276 and counts the first year twice.

$$\mathbf{2.1} \quad S_n = \frac{1(10^n - 1)}{10 - 1} = \boxed{\frac{10^n - 1}{9}}.$$

$\mathbf{2.2} \quad \sum_{i=1}^n S_i = \frac{1}{9} \left(\sum 10^i - \sum 1 \right)$, and the first of those is geometric with $u_1 = 10$, $r = 10$:

$$\sum_{i=1}^n 10^i = \boxed{\frac{10(10^n - 1)}{9}}$$

so the whole thing is $\frac{1}{9} \left(\frac{10(10^n - 1)}{9} - n \right) = \frac{10(10^n - 1) - 9n}{81}$. The 81 is two nines, one from each sum.

$\mathbf{2.3 (i)}$ Both sides have ratio $\frac{3}{2}$, so the area has ratio $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$, and $4 \cdot 5 = 20$ is the first area.

$$\mathbf{(ii)} \quad S_{10} = \frac{20 \left(\left(\frac{9}{4}\right)^{10} - 1 \right)}{\frac{9}{4} - 1} = 16 \left(\left(\frac{9}{4}\right)^{10} - 1 \right),$$

and the mean is a tenth of it:

$$\boxed{p = \frac{8}{5}, a = 10}.$$

3.1 $u_1 = \frac{2}{3} \cdot \frac{7}{8} = \frac{7}{12}$ and $r = \frac{7}{8}$, so

$$S_{\infty} = \frac{7/12}{1 - 7/8} = \boxed{\frac{14}{3}}$$

3.2 $|r| = 3 \geq 1$: the terms do not decrease in size, so the series diverges. " $r < 1$ " is not a reason — **-3 is** less than 1. The markscheme wants $|r| \geq 1$, and accepts $|r| > 1$ or $r < -1$.

3.3 (i) $r = \frac{1.46729}{1.80645} = 0.812252 \dots = \boxed{0.812}$,
and $\frac{1.19181}{1.46729}$ gives the same.

(ii) Each interval is travelled out and back, so the distance in it is $2u_i$:

$$2 \times \frac{1.80645}{1 - 0.812252} = 2(9.62167) = \boxed{19.2 \text{ cm}}$$

3.4 (i) Dividing through by $\ln x$ (legal because $x > 1$, so $\ln x \neq 0$) leaves 1, p , $\frac{1}{3}$:

$$p^2 = \frac{1}{3} \implies \boxed{p = \pm \frac{1}{\sqrt{3}}}$$

(ii) $|p| = \frac{1}{\sqrt{3}} < 1$, so the series converges. Both signs have to be mentioned; considering only one is **R0**.

(iii) With $p = \frac{1}{\sqrt{3}}$,

$$\begin{aligned}
\frac{\ln x}{1 - \frac{1}{\sqrt{3}}} &= 3 + \sqrt{3} \\
\implies \ln x &= \left(3 + \sqrt{3}\right) \left(1 - \frac{1}{\sqrt{3}}\right) \\
&= 3 + \sqrt{3} - \sqrt{3} - 1 \\
&= 2 \\
\boxed{x = e^2}
\end{aligned}$$

4.1 The summand at $i = 1$, not the constant in front:

$$u_1 = \frac{2}{3} \cdot \frac{7}{8} = \boxed{\frac{7}{12}}$$

4.2 This Σ starts at $k = 0$, so here the constant **is** the first term:

$$S_\infty = \frac{53/1000}{1 - 1/100} = \boxed{\frac{53}{990}}$$

4.3 $0.2535353\dots = 0.2 + 0.0535353\dots$:

$$\frac{1}{5} + \frac{53}{990} = \frac{198 + 53}{990} = \boxed{\frac{251}{990}}$$

251 is prime, so nothing cancels.

5.1 $\frac{s}{a} = \frac{t}{s}$ gives $s^2 = at$, that is $\boxed{t = \frac{s^2}{a}}$.

5.2 v_2 and v_4 are two steps apart:

$$5r^2 = 15 \implies r^2 = 3 \implies \boxed{r = \pm\sqrt{3}}$$

$\sqrt{3}$ alone, with no working, scores
M1A1A0.

5.3 $v_{99} = v_2 r^{97}$ and 97 is odd, so $v_{99} < 0$ forces $r = -\sqrt{3}$:

$$v_5 = 5 \left(-\sqrt{3}\right)^3 = \boxed{-15\sqrt{3}}$$

5.4 Equal ratios:

$$\begin{aligned} \frac{3-2k}{k-5} &= \frac{5k+3}{3-2k} \\ &\implies (3-2k)^2 \\ &= (k-5)(5k+3) \end{aligned}$$

$$\begin{aligned} 9-12k+4k^2 &= 5k^2-22k-15 \\ &\implies k^2-10k-24 \\ &= 0 \end{aligned}$$

5.5 $(k-12)(k+2) = 0$, and $k = 12$ was 1.2, so $\boxed{k = -2}$ and the terms are

$$-7, \quad 7, \quad -7$$

with $r = -1$. Substituting into the wrong expression is the standard slip: u_3 is $5k + 3$.

5.6 With $r = -1$ the terms cancel in pairs, and $2m$ terms make m pairs, so $\boxed{S_{2m} = 0}$. No formula needed.

6.1 $50r^3 = 86.4$ gives $r = 1.2$, and

$$\begin{aligned} 250(1.2^n - 1) &> 33\,500 \implies 1.2^n > 135 \\ &\implies n > 26.9045 \dots \\ &\implies \boxed{n = 27} \end{aligned}$$

$S_{26} = 28\,368.8$ and $S_{27} = 34\,092.6$ — check both.

6.2 The critical value is $63.2675\dots$, so $\boxed{n = 64}$; $S_\infty - S_{63} = 0.001036$ is still too big and $S_\infty - S_{64} = 0.000906$ is small enough. Solving $S_\infty - u_n < 0.001$ instead scores **M0**.

6.3 10% of \$35 000 is \$3500:

$$\begin{aligned} 29\,750 \times 0.89^{n-1} &< 3500 \implies n > 19.364 \dots \\ &\implies \boxed{n = 20} \end{aligned}$$

At 19 years the car is worth \$3651.80, at 20 it is \$3250.10.

7.1 The powers run x^0 to x^n , so there are $\boxed{n+1}$ terms, and

$$f(x) = \frac{1(x^{n+1} - 1)}{x - 1} = \boxed{\frac{x^{n+1} - 1}{x - 1}}$$

The **R1** is for the count; using n terms gives $\frac{x^n - 1}{x - 1}$, which is the mistake the mark exists to catch.

7.2 $u_1 = 1$, $r = -x^2$, and $|x| < 1$ makes $|r| < 1$:

$$S_\infty = \frac{1}{1 - (-x^2)} = \boxed{\frac{1}{1 + x^2}}$$

Reading the ratio as x^2 gives $\frac{1}{1-x^2}$ — a different series.

7.3 One more term than f_3 , and $(-2x^2)^4 = 16x^8$:

$$f_4(x) = \boxed{1 - 2x^2 + 4x^4 - 8x^6 + 16x^8}$$

Five terms, because r runs from 0 to 4.